

Unit 4 Geometry

1

★ Mark for Review

ABC

The surface area, in square inches, of a cube can be represented by the expression $24a^2$, where a is a positive constant. Which expression represents the volume, in cubic inches, of the cube?

A. $8a^3$

B. $16a^3$

C. $64a^6$

D. $216a^6$

2

★ Mark for Review

ABC

Given the function $f(x) = \pi x^3 + \pi x^2$, where f represents the volume (in cubic feet) of a right circular cylinder with radius x feet, which expression represents the height (in feet) of the cylinder?

A. $x + 1$

B. πx^3

C. x

D. $x^3 + 1$

3

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ABC

Identical rectangles X and Y each have an area of $\frac{49}{5}$ square meters and a perimeter of P meters. If one of the shorter sides of rectangle X is glued to one of the shorter sides of rectangle Y , the resulting rectangle has a perimeter of $\frac{14}{9}P$ meters. What is the value of P ?

4

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ABC

The hypotenuse of an isosceles right triangle has a length of 12 units. What is the area of the triangle, in square units?

A. $18\sqrt{2}$

B. 36

C. 48

D. $36\sqrt{2}$ **5**

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ABC

In triangles LMN and RST , angles L and R each have measure 127° . Which additional piece of information is sufficient to determine whether triangle LMN is similar to triangle RST ?

A. The measures of angles N and T .B. The lengths of sides LM and RS .C. The measures of angles S and T .D. The lengths of sides MN and ST .

6

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ABC

In triangles LMN and PQR , angles M and Q each have measure 54° , $MN = 16$, and $QR = 16$. Which additional piece of information is sufficient to prove that triangle LMN is congruent to triangle PQR ?

A. Angles L and R each have measure 44° .

B. Angles N and P each have measure 44° .

C. $LM = 11$ and $PQ = 11$.

D. $LN = 11$ and $PR = 11$.

7

★ Mark for Review

ABC

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C. $LM = 11$ and $PQ = 11$.

D. $LN = 11$ and $PR = 11$.

8

★ Mark for Review

ABC

In triangle ABC , the measure of angle A is 25° , angle C is a right angle, and the length of BC is 16. Which of the following represents the length of AC ?

A. $16 \sin 25^\circ$

B. $16 \tan 25^\circ$

C. $\frac{16}{\sin 25^\circ}$

D. $\frac{16}{\tan 25^\circ}$

9

★ Mark for Review

ABC

Triangle JKL is similar to triangle RST , where angle J corresponds to angle R and angles K and S are right angles. If $\cos(L) = \frac{119}{169}$, what is the value of $\tan(T)$?

A. $\frac{120}{169}$

B. $\frac{119}{120}$

C. $\frac{120}{119}$

D. $\frac{169}{119}$

10

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ABC

Point O is the center of a circle. The measure of arc MN on this circle is $\frac{4\pi}{3}$ radians. The length of arc MN is what fraction of the circumference of the circle?

11

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ABC

In the xy -plane, segment OA has endpoints $(0, 0)$ and $(\sqrt{3}, \sqrt{3})$. What is the measure, in radians, of the acute angle formed by segment OA and the line $y = 0$?

A. $\frac{\pi}{6}$

B. $\frac{\pi}{4}$

C. $\frac{\pi}{3}$

D. $\frac{\pi}{2}$

12

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ABC

Triangle ABC is similar to triangle DEF , where angle A corresponds to angle D and angles B and E are right angles. If $AB = 2$, $DF = 12\sqrt{2}$, and $\tan(A) = \sqrt{7}$, what is the length of DE ?

13

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ABC

In the xy -plane, circle A is defined by the equation $(x - 3)^2 + (y - 4)^2 = 16$, and circle B is defined by the equation $(x - 3)^2 + (y - 4)^2 = n$, where n is a constant. The diameter of circle B is 48 units longer than the diameter of circle A . What is the value of n ?

14

★ Mark for Review

ABC

The given equation represents a circle in the xy -plane:

$$x^2 + y^2 + 7x + y = -\frac{7}{2}$$

What is the length of the radius of the circle?

15

★ Mark for Review

ABC

In the xy -plane, a circle has center D with coordinates $(m, 16)$, where m is a constant. Points E and F lie on the circle. Point E lies on the circle and has coordinates $(m+4\sqrt{10}, 34)$, and point F has coordinates $(n, 16)$, where n is a constant greater than m . What is the area, in square units, of triangle DEF ?

A. 198

B. $72\sqrt{10}$

C. 242

D. $99\sqrt{10}$ **16**

★ Mark for Review

ABC

In the xy -plane, a circle has center D with coordinates $(m, 16)$, where m is a constant. Points E and F lie on the circle. Point E lies on the circle and has coordinates $(m+4\sqrt{10}, 34)$, and point F has coordinates $(n, 16)$, where n is a constant greater than m . What is the area, in square units, of triangle DEF ?

A. 198

B. $72\sqrt{10}$

C. 242

D. $99\sqrt{10}$

17

★ Mark for Review

ABC

A circle in the xy -plane has the equation:

$$(x - 2)^2 + (y - 2)^2 = 25.$$

Line m is tangent to this circle at the point $(5, 6)$. What is the y -coordinate of the y -intercept of line m ?

18

★ Mark for Review

ABC

If $\theta = 3x + 4$, and $\beta = 6x - 5$, What is the value of x so that $\sin(\theta) = \cos(\beta)$?

19

★ Mark for Review

ABC

A circle has center O , and points M and N lie on the circle. The measure of arc MN is 24° and the length of arc MN is 8π centimeters. What is the length, in centimeters, of the radius of the circle?

A. 15

B. 45

-C. 60

D. 120

20

★ Mark for Review

ABC

Which of the following equations represents a circle in the xy -plane that intersects the line $x = 5$ at exactly one point?

A. $(x - 1)^2 + (y - 7)^2 = 4$

B. $(x - 5)^2 + (y - 1)^2 = 4$

C. $(x - 7)^2 + (y - 5)^2 = 4$

D. $(x - 13)^2 + (y - 13)^2 = 4$

21

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ABC

A circle has center A , and points B and C lie on the circle. Line segment BD is tangent to the circle at point B , line segment CD is tangent to the circle at point C , and the length of line segment BD is 63 units. If the circle has a radius of 16 units, what is the area, in square units, of quadrilateral $ABCD$?

22

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ABC

In the xy -plane, a circle with a radius of 1 has its center at $(9, -12)$. This circle can be represented by the equation $x^2 + y^2 + ax + by + c = 0$, where a , b , and c are constants. What is the value of $a + b + c$?

SAT Geometry Unit 4

Complete Step-by-Step Answer Key

Novel Prep Learning Center

Quick Answer Key

1	A	2	A	3	$\frac{63}{5}$
4	B	5	A	6	C
7	C	8	D	9	C
10	$\frac{2}{3}$	11	B	12	6
13	784	14	3	15	A
16	A	17	$\frac{39}{4}$	18	$\frac{91}{9}$
19	C	20	C	21	1008
22	230				

Question 1

The surface area of a cube is

$$6s^2 = 24a^2$$

Divide both sides by 6:

$$s^2 = 4a^2$$

Since $s > 0$,

$$s = 2a$$

The volume of a cube is

$$V = s^3$$

So

$$V = (2a)^3 = 8a^3$$

$$\boxed{8a^3}$$

Question 2

The volume of a cylinder is

$$V = \pi r^2 h$$

Here the radius is x , so

$$V = \pi x^2 h$$

Given

$$f(x) = \pi x^3 + \pi x^2$$

Factor:

$$f(x) = \pi x^2(x + 1)$$

Therefore,

$$h = x + 1$$

$$\boxed{x + 1}$$

Question 3

Let the rectangle have shorter side s and longer side l .

Area:

$$sl = \frac{49}{5}$$

Original perimeter:

$$P = 2l + 2s = 2(l + s)$$

When the two rectangles are glued along their shorter sides, the new rectangle has dimensions

$$2l \times s$$

So the new perimeter is

$$2(2l + s)$$

The problem says

$$2(2l + s) = \frac{14}{9}P$$

Since $P = 2(l + s)$,

$$2(2l + s) = \frac{14}{9} \cdot 2(l + s)$$

Divide both sides by 2:

$$2l + s = \frac{14}{9}(l + s)$$

Multiply by 9:

$$18l + 9s = 14l + 14s$$

$$4l = 5s$$

$$l = \frac{5}{4}s$$

Use the area:

$$sl = \frac{49}{5}$$

$$s \left(\frac{5}{4}s \right) = \frac{49}{5}$$

$$\frac{5}{4}s^2 = \frac{49}{5}$$

Multiply by 4/5:

$$s^2 = \frac{196}{25}$$

$$s = \frac{14}{5}$$

Then

$$l = \frac{5}{4} \cdot \frac{14}{5} = \frac{7}{2}$$

So

$$P = 2(l + s)$$

$$P = 2 \left(\frac{7}{2} + \frac{14}{5} \right)$$

$$P = 2 \left(\frac{35}{10} + \frac{28}{10} \right)$$

$$P = 2 \cdot \frac{63}{10}$$

$$\boxed{\frac{63}{5}}$$

Question 4

An isosceles right triangle is a $45^\circ - 45^\circ - 90^\circ$ triangle.

If each leg is x , then the hypotenuse is

$$x\sqrt{2}$$

Given hypotenuse 12:

$$x\sqrt{2} = 12$$

$$x = \frac{12}{\sqrt{2}} = 6\sqrt{2}$$

Area:

$$A = \frac{1}{2}x^2$$

$$A = \frac{1}{2}(6\sqrt{2})^2$$

$$A = \frac{1}{2}(72) = 36$$

$$\boxed{36}$$

Question 5

We already know

$$\angle L = \angle R = 127^\circ$$

To prove two triangles are similar, we need two pairs of congruent angles.

If we know the measures of angles N and T , then we can compare another angle pair. Once two angle pairs match, the triangles are similar by AA similarity.

$$\boxed{A}$$

Question 6

Given:

$$\angle M = \angle Q = 54^\circ$$

and

$$MN = QR = 16$$

To prove congruence, we need enough information for SAS, ASA, AAS, or SSS.
If

$$LM = PQ = 11$$

then we have two sides and the included angle:

$$LM \leftrightarrow PQ, \quad MN \leftrightarrow QR, \quad \angle M \leftrightarrow \angle Q$$

This proves congruence by SAS.

□ C

Question 7

This is the same as Question 6.

The sufficient information is

$$LM = 11 \quad \text{and} \quad PQ = 11$$

because this gives SAS congruence.

□ C

Question 8

In right triangle ABC ,

$$\angle A = 25^\circ$$

and $BC = 16$.

Relative to angle A ,

$$BC = \text{opposite}$$

and

$$AC = \text{adjacent}$$

So

$$\tan 25^\circ = \frac{BC}{AC}$$

$$\tan 25^\circ = \frac{16}{AC}$$

Solve for AC :

$$AC = \frac{16}{\tan 25^\circ}$$

$$\boxed{\frac{16}{\tan 25^\circ}}$$

Question 9

Since $\triangle JKL \sim \triangle RST$,

$$L \leftrightarrow T$$

Given

$$\cos L = \frac{119}{169}$$

In a right triangle,

$$\cos L = \frac{\text{adjacent}}{\text{hypotenuse}}$$

So adjacent = 119, hypotenuse = 169.

Find the opposite side:

$$169^2 - 119^2$$

$$= 28561 - 14161 = 14400$$

$$\sqrt{14400} = 120$$

Thus,

$$\tan L = \frac{120}{119}$$

Since $L \leftrightarrow T$,

$$\tan T = \frac{120}{119}$$

$$\boxed{\frac{120}{119}}$$

Question 10

The measure of the arc is

$$\frac{4\pi}{3}$$

The full circumference corresponds to

$$2\pi$$

So the fraction of the circumference is

$$\begin{aligned} & \frac{\frac{4\pi}{3}}{2\pi} \\ &= \frac{4\pi}{3} \cdot \frac{1}{2\pi} \\ &= \frac{4}{6} \end{aligned}$$

$$\boxed{\frac{2}{3}}$$

Question 11

The segment has endpoints

$$(0, 0)$$

and

$$(\sqrt{3}, \sqrt{3})$$

The slope is

$$\frac{\sqrt{3}}{\sqrt{3}} = 1$$

So the angle with the positive x -axis satisfies

$$\tan \theta = 1$$

Therefore,

$$\theta = \frac{\pi}{4}$$

$$\boxed{\frac{\pi}{4}}$$

Question 12

Triangle ABC is right at B .

Given

$$AB = 2$$

and

$$\tan A = \sqrt{7}$$

Since

$$\tan A = \frac{BC}{AB}$$

we get

$$\sqrt{7} = \frac{BC}{2}$$

$$BC = 2\sqrt{7}$$

Now find hypotenuse AC :

$$AC^2 = AB^2 + BC^2$$

$$AC^2 = 2^2 + (2\sqrt{7})^2$$

$$AC^2 = 4 + 28 = 32$$

$$AC = 4\sqrt{2}$$

Since triangle $ABC \sim DEF$, and $AC \leftrightarrow DF$,

$$AC = 4\sqrt{2}, \quad DF = 12\sqrt{2}$$

Scale factor:

$$\frac{12\sqrt{2}}{4\sqrt{2}} = 3$$

Since $AB \leftrightarrow DE$,

$$DE = 2 \cdot 3 = 6$$

$$\boxed{6}$$

Question 13

Circle A :

$$(x - 3)^2 + (y - 4)^2 = 16$$

So

$$r_A = 4$$

Diameter of circle A :

$$d_A = 8$$

Circle B has diameter 48 units longer:

$$d_B = 8 + 48 = 56$$

So

$$r_B = 28$$

The equation of circle B is

$$(x - 3)^2 + (y - 4)^2 = n$$

where

$$n = r_B^2$$

$$n = 28^2 = 784$$

$$\boxed{784}$$

Question 14

Given

$$x^2 + y^2 + 7x + y = -\frac{7}{2}$$

Group terms:

$$x^2 + 7x + y^2 + y = -\frac{7}{2}$$

Complete the square.

For $x^2 + 7x$, add

$$\left(\frac{7}{2}\right)^2 = \frac{49}{4}$$

For $y^2 + y$, add

$$\left(\frac{1}{2}\right)^2 = \frac{1}{4}$$

So

$$x^2 + 7x + \frac{49}{4} + y^2 + y + \frac{1}{4} = -\frac{7}{2} + \frac{49}{4} + \frac{1}{4}$$

$$\left(x + \frac{7}{2}\right)^2 + \left(y + \frac{1}{2}\right)^2 = -\frac{14}{4} + \frac{50}{4}$$

$$\left(x + \frac{7}{2}\right)^2 + \left(y + \frac{1}{2}\right)^2 = 9$$

So

$$r^2 = 9$$

$$\boxed{3}$$

Question 15

Center:

$$D = (m, 16)$$

Point

$$E = (m + 4\sqrt{10}, 34)$$

The radius is DE .

Horizontal distance:

$$4\sqrt{10}$$

Vertical distance:

$$34 - 16 = 18$$

So

$$DE^2 = (4\sqrt{10})^2 + 18^2$$

$$DE^2 = 160 + 324 = 484$$

$$DE = 22$$

Point $F = (n, 16)$ is on the circle and to the right of D , so

$$DF = 22$$

Triangle DEF has base

$$DF = 22$$

and height

$$34 - 16 = 18$$

Area:

$$A = \frac{1}{2}(22)(18)$$

$$A = 198$$

$$\boxed{198}$$

Question 16

This is the same as Question 15.

$$DE = 22, \quad DF = 22, \quad \text{height} = 18$$

$$A = \frac{1}{2}(22)(18) = 198$$

$$\boxed{198}$$

Question 17

Circle:

$$(x - 2)^2 + (y - 2)^2 = 25$$

Center:

$$(2, 2)$$

Tangent point:

$$(5, 6)$$

Slope of radius:

$$m_r = \frac{6-2}{5-2} = \frac{4}{3}$$

The tangent line is perpendicular to the radius, so its slope is the negative reciprocal:

$$m_t = -\frac{3}{4}$$

Use point-slope form:

$$y - 6 = -\frac{3}{4}(x - 5)$$

Find the y -intercept by setting $x = 0$:

$$y - 6 = -\frac{3}{4}(0 - 5)$$

$$y - 6 = \frac{15}{4}$$

$$y = 6 + \frac{15}{4}$$

$$y = \frac{24}{4} + \frac{15}{4}$$

$$\boxed{\frac{39}{4}}$$

Question 18

We need

$$\sin(\theta) = \cos(\beta)$$

For complementary angles,

$$\sin(\theta) = \cos(90^\circ - \theta)$$

So we set

$$\theta + \beta = 90^\circ$$

Given

$$\theta = 3x + 4$$

and

$$\beta = 6x - 5$$

Then

$$3x + 4 + 6x - 5 = 90$$

$$9x - 1 = 90$$

$$9x = 91$$

$$\boxed{\frac{91}{9}}$$

Question 19

Arc length formula:

$$s = r\theta$$

Convert 24° to radians:

$$24^\circ = \frac{24\pi}{180} = \frac{2\pi}{15}$$

Given arc length:

$$s = 8\pi$$

So

$$8\pi = r \cdot \frac{2\pi}{15}$$

Divide by π :

$$8 = \frac{2r}{15}$$

$$120 = 2r$$

$$r = 60$$

$$\boxed{60}$$

Question 20

A circle intersects the vertical line $x = 5$ at exactly one point if the line is tangent to the circle.

So the distance from the center to the line $x = 5$ must equal the radius.

Check choice C:

$$(x - 7)^2 + (y - 5)^2 = 4$$

Center:

$$(7, 5)$$

Radius:

$$2$$

Distance from center to line $x = 5$:

$$|7 - 5| = 2$$

Distance equals radius, so the line is tangent.

$$\boxed{C}$$

Question 21

Tangents from the same external point are congruent:

$$BD = CD = 63$$

The radius is perpendicular to the tangent, so

$$AB \perp BD$$

and

$$AC \perp CD$$

Thus quadrilateral $ABCD$ is made of two right triangles:

$$\triangle ABD$$

and

$$\triangle ACD$$

Each has legs 16 and 63.

Area of one triangle:

$$\frac{1}{2}(16)(63) = 504$$

Area of both triangles:

$$504 + 504 = 1008$$

$$\boxed{1008}$$

Question 22

Circle center:

$$(9, -12)$$

Radius:

$$1$$

Standard form:

$$(x - 9)^2 + (y + 12)^2 = 1$$

Expand:

$$x^2 - 18x + 81 + y^2 + 24y + 144 = 1$$

$$x^2 + y^2 - 18x + 24y + 225 = 1$$

Move 1 to the left:

$$x^2 + y^2 - 18x + 24y + 224 = 0$$

Compare with

$$x^2 + y^2 + ax + by + c = 0$$

So

$$a = -18, \quad b = 24, \quad c = 224$$

Then

$$a + b + c = -18 + 24 + 224$$

$$\boxed{230}$$